ISP Network Engineering Lab

OSPF Area 0 | Fault Isolation | Incident Recovery | Cisco Packet Tracer

Technical incident report • Prepared for Network Engineer interview follow-up • November 2026

Executive summary

This personal Cisco Packet Tracer exercise models a small service-provider backbone with three provider routers and two customer-edge devices. Its purpose is to demonstrate applied OSPF configuration, verification, troubleshooting and recovery after a simulated router-interface outage. This is a controlled simulation, not production network experience.

1. Architecture and addressing

Logical topology: CE1 — PE1 — P1 — PE2 — CE2. OSPF Area 0 runs on PE1, P1 and PE2. CE1 and CE2 are outside this provider OSPF domain.

Connection	Left endpoint	Right endpoint
CE1–PE1	CE1 Gi0/0: 192.0.2.1/30	PE1 Gi0/0: 192.0.2.2/30
PE1–P1	PE1 Gi0/1: 10.0.12.1/30	P1 Gi0/0: 10.0.12.2/30
P1–PE2	P1 Gi0/1: 10.0.23.1/30	PE2 Gi0/1: 10.0.23.2/30
PE2–CE2	PE2 Gi0/0: 192.0.2.5/30	CE2 Gi0/0: 192.0.2.6/30

Provider loopbacks: PE1 1.1.1.1/32; P1 2.2.2.2/32; PE2 3.3.3.3/32.

2. Routing implementation

OSPF process 1 was configured within Area 0 across the provider-facing 10.0.12.0/30 and 10.0.23.0/30 links. Loopback /32 addresses were advertised, and loopback interfaces were made passive. Explicit OSPF router IDs correspond to the loopbacks.

Verified in earlier lab screenshots and user-reported checks: PE1 learned OSPF routes to 2.2.2.2/32, 3.3.3.3/32 and 10.0.23.0/30; P1 displayed two FULL OSPF neighbors before the controlled fault; PE1 reached 3.3.3.3 with successful ping replies.

3. Incident: provider link unavailable

Field	Description
Incident reference	LAB-INC-001 (portfolio identifier)
Scenario	Controlled administrative shutdown on P1 Gi0/1 (link to PE2)
Symptoms	PE2 OSPF neighbor absent on P1; remote loopback expected unreachable from PE1
Immediate cause	P1 Gi0/1 disabled by shutdown command
Scope	Provider backbone path to PE2 in a nonredundant linear topology

4. Troubleshooting and recovery method

Sequence	Command or action	Technical purpose

1	PE1: ping 3.3.3.3	Confirm remote destination reachability
2	PE1: show ip route 3.3.3.3	Inspect route to the affected loopback
3	PE1: ping 10.0.12.2; ping 2.2.2.2	Isolate functional first-hop connectivity
4	P1: show ip interface brief	Check whether the provider-facing port is enabled
5	P1: show ip ospf neighbor	Examine loss of OSPF adjacency to PE2
6	P1: show running-config	Confirm administrative shutdown in interface configuration
7	P1: interface GigabitEthernet0/1; no shutdown	Restore the intended link configuration
8	P1: show ip ospf neighbor; PE1: ping 3.3.3.3	Verify routing recovery and end-to-end reachability

5. Evidence status

Previously reviewed screenshots confirm P1 Gi0/1 as administratively down/down and only the PE1 OSPF neighbor remaining FULL during the simulated outage. The project owner reported successful initial OSPF verification, end-to-end ping, and completion of a video showing investigation and repair. This report does not represent a direct machine-verified extraction of the uploaded .pkt configuration, nor a frame-by-frame validation of every recovery output. The attached project should be opened in Cisco Packet Tracer for independent configuration inspection.

6. Key lessons and production considerations

A successful network repair requires more than issuing a command: establish the failure scope, inspect routing and interfaces, verify the likely root cause, apply a controlled correction, and repeat the original test. In a production ISP, confirm authorization/change history before re-enabling a shut interface. Also, OSPF cannot route around a failed link without a physically available alternative path.

7. Portfolio files and recommended next experiment

Project: ISP_Network_Engineering_Lab.pkt. Recording: OSPF outage and recovery walkthrough. Next recommended exercise: add a redundant PE1–PE2 interface connection, adjust OSPF interface costs, observe path selection, and document automatic failover. Keep the final video and configuration file linked in a GitHub repository with a readable project overview.

Prepared as a transparent account of a simulated learning project following technical interview feedback.